CH 2 - Probability Rules

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2.1 Basic Axioms and Laws

📘 **Definition — Axioms of Probability**

1. The probability of the sample space occurring is $P(S) = 1$
2. Each event of happing in the sample space is $0 \leq P(A) \leq 1$
3. If $A_1, A_2, \ldots$ have no overlap, then $P(A_1 \cup A_2 \cup \ldots) = \sum_i P(A_i)$

Note:

1. A "or" B means $A \cup B$
2. A "and" B means $A \cap B$
3. "Not in" A means the compliment $\overline{A}$

📘 **Definition — De Morgan's Law**

1. $\overline{A \cup B} = \overline{A} \cap \overline{B}$
2. $\overline{A \cap B} = \overline{A} \cup \overline{B}$

2.2 Probability Rules

1. If $A \subseteq B$ then $P(A) \leq P(B)$
2. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
3. $P(A \cup B \cup C) = P(A \cup B) + P(C) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$
4. In general, $P(A_1 \cup \ldots \cup A_k) = \sum_i^k P(A_i) - \sum\left(\sum_{j \neq i} P(A_i A_j)\right) + \sum\left(\sum\left(\sum_{i \neq j \neq k} P(A_i A_j A_k)\right)\right) + \ldots - \ldots$

Proof

$B = (B \cap A) \cup \left(B \cap \overline{A}\right)$

$\implies P(B) = P(B \cap A) + P\left(B \cap \overline{A}\right) = P(A) + W,\ W > 0 \implies P(B) > P(A)$

Note: $P(B) = P(AB) + P\left(\overline{A}B\right)$

📘 **Definition — Mutual Exclusive**

Event A and B are **mutually exclusive** if $A \cap B = \varnothing$. Non-overlapping, disjoint means the same.

If $A_1, ..., A_k$ are mutually exclusive, $P(A_1 \cup ... \cup A_k) = \sum_i^k P(A_i)$

Examples:

Roll two fair 12-sided dice. What is the probability at least one of them is larger than 7?

$1 - \frac{7}{12} \cdot \frac{7}{12} = \frac{95}{144}$ or $\frac{5}{12} \cdot \frac{7}{12} \cdot 2 + \left(\frac{5}{12}\right)^2 = \frac{95}{144}$

2.3 Independent Events

📘 **Definition — Independent Events**

Two events A and B are independent if and only if

$$P(AB) = P(A)P(B)$$

If we can directly compute $P(A), P(B), P(AB)$, we can determine whetheher or not the events are independent.

For multiple events, $A_1, A_2, ..., A_k$, we say they are independent if and only if

$$P(A_iA_j) = P(A_i)P(A_j)$$
$$P(A_iA_jA_k) = P(A_i)P(A_j)P(A_k)$$
$$...$$

Example:

1. Roll 2 fair 6-sided dice. Let A =first die shows 3, B = sum of dice is 7

   $P(A) = \frac{1}{6}, P(B) = \frac{1}{6}$

   $P(AB) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$

2. What about sum is 8

   Let C =total on 2 dice is 8

   $P(C) = \frac{5}{36}$

   $P(A \cap C) = \frac{1}{36} \neq \frac{1}{6}\frac{5}{36}$. Thus C and A are not independent

Note: If A and B are linearly independent, so $\overline{A} \wedge B$, $A \wedge \overline{B}$, $\overline{A} \wedge \overline{B}$ are also linearly independent.

Summary:

	Mutually Exclusive	Indepedent	Both
Words	No overlap	No relation	At least one event is impossible
Math	$A \cap B = \varnothing$	$P(AB) = P(A)P(B)$	$P(A) = 0 \vee P(B) = 0$

📘 **Definition — Conditional Probability**

The conditional probability, meaning the probability of A given B denoted as

$$P(A \mid B) = \frac{P(AB)}{P(B)}, \quad P(B) > 0$$

If A and B are indepedent events,

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

which can be interpreted as another definition of independence.

Example:

Roll 2 fair 6-sided dice.

$A = \text{first is } 3, B = \text{total } 7, C = \text{total } 8$

$P(A) = \frac{1}{6}, P(C) = \frac{5}{36}, P(AC) = \frac{1}{36}$

$P(A|C) = \frac{\frac{1}{36}}{\frac{5}{36}} = \frac{1}{5}$

$C = \{(2,6),(3,5),(4,4),(5,3),(6,2)\}$, only $(3,5)$ satisfy A out of 5 choices in the sample space, thus $\frac{1}{5}$

$P(C|A) = \frac{P(AC)}{P(A)} = \frac{\frac{1}{36}}{\frac{1}{6}} = \frac{1}{6}$

$A = \{(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)\}$, only $(3,5)$ satisfy C out of 6 choices in the sample space, thus $\frac{1}{6}$

💡 **Tip** — Dependence and Conditional Probability

Dependence is always a two-way street.

$$P(A|B) > P(A) \iff P(B|A) > P(B)$$
$$P(A|B) = P(A) \iff P(B|A) = P(B)$$
$$P(A|B) < P(A) \iff P(B|A) < P(B)$$

If $P(A|B) < P(A) \implies P\left(\overline{A}|B\right) > P\left(\overline{A}\right)$

📘 **Definition — Product Rule for Intersections**

For any two events $P(A|B) = \frac{P(AB)}{P(B)}, P(B|A) = \frac{P(AB)}{P(A)}$

so

$$P(AB) = P(B)P(A|B)$$
$$P(AB) = P(A)P(B|A)$$

For any 3 events

$$\begin{aligned} P(ABC) &= P(A)P(B|A)P(C|AB) \\ &= P(C)P(C|A)P(B|AC) \\ &= \dots \end{aligned}$$

Examples:

1. Suppose each child has 50% chance of having red hair, independently. At least one of the kids has red hair, what is the probability of the other child has.

	R	N
R	0.25	0.25
N	0.25	0.25

   Given that at least one has red hair, then our sample space actually becomes {RN, RR, NR} and the probability becomes 1/3

📘 **Definition — Law of Total Proabability**

A partition is a collection of event A where $A_i \cap A_j = \varnothing, \forall i, j.\ A_1 \cup \dots \cup A_k = S$

Suppose we have a partition $A_1, \dots, A_k$ and any event B, then $P(B) = \sum_{i=1}^{k} P(B \cap A_i) = \sum_{i=1}^{k} P(A_i)P(B|A_i)$

Example:

1. Suppose a blood test has 5% false positive rate and 1% false negative rate. And 2% of a populaiton has disease. What is the probability a random person is positive.

   Let D be event person has disease and T be event person is positive

   $P(D) = 0.02, P\left(T|\overline{D}\right) = 0.05, P\left(\overline{T}|D\right) = 0.01$

   $P(T) = P(T|D)P(D) + P\left(\overline{D}\right)P\left(T|\overline{D}\right) = 0.02(1 - 0.01) + (1 - 0.02)(0.05) = 0.0688$

2. Three AI are contributing to a project.

   A contributes twice as many as B or C. 10% of A has error, 20% of B has error, 50% of C has error

   Let A, B, C be the event of A, B, C contribution respectively. Let E be error.

   $P(A) = 0.5, P(B) = 0.25, P(C) = 0.25$

   $P(E|A) = 0.1, P(E|B) = 0.2, P(E|C) = 0.5$

   $P(E) = 0.5 \cdot 0.1 + 0.25 \cdot 0.2 + 0.25 \cdot 0.5 = 0.225$

We might often want to reveerse the direction of conditioning. For an example, the previous example, we want to know who wrote it which we an use **Bayes Law**

📘 **Definition — Bayes' Law**

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)P(B|A)}{P(B)} = \frac{P(A)P(B|A)}{P(A)P(B|A) + P\left(\overline{A}\right)P\left(B|\overline{A}\right)}$$

Example:

1. Continuation of previous example on disease testing, given a positive test, what probability the person has disease?

   $P(D|T) = \frac{P(D)P(T|D)}{P(T)} = \frac{0.02 \cdot 0.99}{0.0688} \approx 0.2877$

2. Continuation of example on coding,

   $P(A|E) = \frac{P(A)P(E|A)}{P(E)} = \frac{0.5 \cdot 0.1}{0.225} \approx 0.222$

   $P(B|E) \approx 0.222, P(C|E) \approx 0.556$

   Notice $P(A|E) + P(B|E) + P(C|E) = 1$ as they are a partition.